



Structure & Syllabus for Semester-V
Integrated M.Sc. (Computer Science) Programme
offered in
Department of Computer Science
Gujarat University
2025 – 2026

**As per NEP 2020 CURRICULUM AND CREDIT FRAMEWORK FOR
UNDERGRADUATE PROGRAMMES, UGC**

&

Resolution No. KCG/admin/2023-24/0607/kh.1

of

Education Department, Govt. of Gujarat

Semester - V

(Integrated M.Sc. (Computer Science) Programme)

MINORS

- 1. ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING (AIML)**
- 2. INFORMATION SECURITY (IS)**
- 3. WEB TECHNOLOGIES (WT)**

Course Name: Data Analytics

Course Code: DSC-M-CS-354T

Credits: 2

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Introduce the necessary understanding of human intelligence and to explore the mechanisms that enables the intelligent thought and action
- **CO2:** Understand and learn effective ways for representing knowledge, applying intelligent problem-solving techniques & searching techniques

Prerequisites:

Basic knowledge of Python

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Introduction to Data Scales, Various Data Measures, Probability and Distributions: <i>Introduction to Data Scales and Measures of Location:</i> Mean, Median, Mode, Percentiles, Quartiles; <i>Measures of Variability:</i> Range, Interquartile range, Variance, Standard deviation, Coefficient of variation; <i>Measures of Association between two variables:</i> Definition of Correlation, Types of Correlation, Methods of Correlation, Covariance interpretation of the covariance correlation, Coefficient interpretation of the correlation coefficient; <i>Measures of Distribution:</i> Shape, Relative location, and detecting outliers z-scores empirical rule detecting outliers box plot <i>Introduction to Probability:</i> Counting rules, Combinations and Permutations assigning probabilities, Events and their probabilities, Some basic relationships of probability, Complement of an event addition law, Conditional probability, Independent events multiplication law; <i>Probability Distributions:</i> Random Variables, Discrete vs Continuous random variable, <i>Discrete probability distribution:</i> Binomial, Poisson, Hyper geometric, <i>Continuous probability distribution:</i> Normal and standard normal distributions	01	15 Hrs.
2.	Unit Title: Sampling, Sampling Distribution, Interval Estimation, Hypothesis Testing and Overview of Non-Parametric tests: <i>Sampling, Sampling Distribution:</i> Random Sample Techniques, Non-random Sampling, Sampling and Non-sampling Errors, Sampling Distribution of Means, Sampling Distribution of Population, Properties of Point Estimators, <i>Interval Estimation:</i> Population Mean (σ : known & unknown), Determining Sample Size, Population Proportion <i>Hypothesis Test:</i> Null hypothesis, Alternative hypothesis, Simple and	01	15 Hrs.

Unit No.	Particulars	Credits	Time
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composite hypothesis, Type- I & type-II Errors, Test of significance of population Mean (σ : known & unknown), Level of significance, Determining test statistics, p-value approach, Critical value approach Determination of Sample size - (z-test, t-test), Overview of Non-Parametric tests

References:

1. Anderson, Sweeney, Williams, "Statistics for Business and Economics", 12th edition, Cengage Publishers
2. Black, "Applied Business Statistics", 7th edition, Wiley
3. Aczel, Sounderpandian, Saravanan, Joshi, "Complete Business Statistics", 7th edition, McGraw-Hill
4. Levin, Rubin, "Statistics for Management", 8th edition, Pearson
5. David P. Doane, Lori E. Seward, "Applied Statistics in Business and Economics", 3rd edition, McGraw-Hill
6. Larry Wasserman, "All of Statistics", Springer
7. Richard A. Johnson, "Probability and Statistics for Engineers", 6th edition, PHI

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Develop strong analytical skills by working with real datasets
- Perform statistical tests (z-test, t-test), and interpreting results using software tools
- Derive meaningful insight through performing data visualization, probability applications, and hypothesis validation
- Apply statistical methods accurately in professional setups

Course Name: Data Analytics – Practicals

Course Code: DSC-M-CS-354P

Credits: 2

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Understand key concepts of data, data mining, data analysis and data analytics
- **CO2:** Understand how data is created, stored, accessed
- **CO3:** Understand tools to effectively organize and visualize data
- **CO4:** Put up the principles and methods of statistical analysis into practice using a range of real-world data sets

Prerequisites:

Basic of Python

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Implementation of Various Data Measures, Probability Rules and Distributions <i>Measures of Central Tendency, Measure of Dispersion, Measures of Association between two variables:</i> Perform computation programmatically, <i>Probability Concepts:</i> Demonstration of basic probability rules using random and collections, Basic relationships of probability, Conditional probability, <i>Probability Distributions:</i> Performing Discrete probability distributions and Continuous probability distributions programmatically	01	30 Hrs.
2.	Unit Title: Computation of Interval Estimation and Hypothesis Testing <i>Interval Estimation:</i> Population Mean (σ : known & unknown), <i>Hypothesis Test:</i> Test of significance of population Mean (σ : known & unknown)	01	30 Hrs.

References:

1. Anderson, Sweeney, Williams, "Statistics for Business and Economics", 12th edition, Cengage Publishers
2. Black, "Applied Business Statistics", 7th edition, Wiley
3. Aczel, Sounderpandian, Saravanan, Joshi, "Complete Business Statistics", 7th edition, McGraw-Hill
4. Levin, Rubin, "Statistics for Management", 8th edition, Pearson
5. David P. Doane, Lori E. Seward, "Applied Statistics in Business and Economics", 3rd edition, McGraw-Hill
6. Larry Wasserman, "All of Statistics", Springer
7. Richard A. Johnson, "Probability and Statistics for Engineers", 6th edition, PHI

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

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- Perform statistical tests (z-test, t-test), and interpreting results using software tools
- Derive meaningful insight through performing data visualization, probability applications, and hypothesis validation
- Apply statistical methods accurately in professional setups

Course Name: Artificial Intelligence

Course Code: DSC-M-CS-355T

Credits: 2

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Introduce the necessary understanding of human intelligence and explore the mechanisms that enable the intelligent thought and action
- **CO2:** Understand and learn effective ways for representing knowledge, applying intelligent problem-solving techniques & searching techniques

Prerequisites:

Basic knowledge of Mathematical Logic

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: AI Fundamentals, State Space Search and Heuristic Search Techniques: <i>Defining Artificial Intelligence (AI):</i> Early history, Overview of AI Application Areas, Defining AI techniques, Turing Test, Intelligent Agents, Defining problems as State Space Search, Production Systems and Characteristics, Breadth First and Depth First Search, Hill Climbing Techniques, Problems and Solutions, Best First Search, A* Algorithm, AO* Algorithm, Iterative Deepening A*, Game Playing: Overview, Applications, Minimax Method, Alpha Beta Pruning	01	15 Hrs.
2.	Unit Title: Knowledge Representation Knowledge Representation Techniques: Semantic Nets, Partitioned Semantic Nets, Conceptual Dependency, Rules for Knowledge Representation, Advantages, Limitations, Representing simple facts in logic, Propositional Logic, FOPL, Computable functions and predicates, Procedural vs. Declarative knowledge, Forward Vs. Backward Chaining	01	15 Hrs.

References:

1. Elaine Rich, Kevin Knight, Shivashankar B. Nair, "Artificial Intelligence", 3rd Edition, McGraw Hill
2. Stuart Russell and Peter Norvig "Artificial Intelligence-A modern Approach", 2nd Edition, Pearson Education
3. George Luger, "Artificial Intelligence - Structures and Strategies for Complex Problem Solving", Pearson Education
4. Michael Negnevitsky, "Artificial Intelligence - A guide to Intelligent Systems", Pearson Education
5. N.P. Padhy, "Artificial Intelligence and Intelligent Systems", Oxford University Press

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Store and represent the knowledge in various applications and use different AI searching techniques
- Deal with poorly defined or inexact problems that do not respond to the algorithmic solutions

Course Name: Artificial Intelligence – Practicals

Course Code: DSC-M-CS-355P

Credits: 2

Course Outcomes:

The aim of this course is to enable students to

- **CO1:** Implement the necessary algorithms related to problem solving and game playing using suitable programming languages
- **CO2:** Analyse the search techniques and choosing the best alternative based on the given task

Prerequisites:

Basic knowledge of Mathematical logic and programming

Contents:

Unit No.	Particulars	Credits	Time
1.	Unit Title: Implementation of AI necessary algorithms and games Tic – Tac – Toe, 3 – Water Jug, Tower of Hanoi, 8 Puzzle problem	01	30 Hrs.
2.	Unit Title: Search Techniques Breadth First Search, Depth First Search, Best First Search, A* Algorithm, AO* Algorithm	01	30 Hrs.

References:

1. Elaine Rich, Kevin Knight, Shivashankar B. Nair, "Artificial Intelligence", 3rd Edition, McGraw Hill
2. Stuart Russell and Peter Norvig "Artificial Intelligence-A modern Approach", 2nd Edition, Pearson Education
3. George Luger, "Artificial Intelligence - Structures and Strategies for Complex Problem Solving", Pearson Education
4. Michael Negnevitsky, "Artificial Intelligence - A guide to Intelligent Systems", Pearson Education
5. N.P. Padhy, "Artificial Intelligence and Intelligent Systems", Oxford University Press

Accomplishments of the student after completing the course:

At the end of the course, students will be able to

- Practically implement Artificial intelligence concepts such as search algorithms and gaming algorithms
